# Distinct rotations send the representative set to disjoint sets

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*Editorial scope.* Statement and complete author proof are included verbatim as the span lines around ollabel vitalinooverlap of the pinned original source; no text is rewritten or paraphrased.

*Allowed standard prerequisites (editorial).*

- the rotation group acting on the sphere
- the representative set C selecting one member per equivalence class of the orbit relation
- composition and inverses of rotations staying in the group

**Lemma 0.1.** *If* $\rho_1 \neq \rho_2$ *then* $\mathrm{im}(\rho_1)C \cap \mathrm{im}(\rho_2)C = \emptyset$.

*Proof.* Suppose $s \in \mathrm{im}(\rho_1)C \cap \mathrm{im}(\rho_2)C$. So $s = \rho_1(r_1) = \rho_2(r_2)$ for some $r_1, r_2 \in C$. Hence $\rho_2^{-1}(\rho_1(r_1)) = r_2$, and $\circ\rho_1\rho_2^{-1} \in \mathrm{SO}(2)$, so $r_1 \sim r_2$. So $r_1 = r_2$, as $C$ selects exactly one member from each equivalence class under $\sim$. So $s = \rho_1(r_1) = \rho_2(r_1)$, and hence $\rho_1 = \rho_2$. □

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